

Introduction to Linear Optimization with Python and Gurobi

1 Optimization

Imagine that you are a mutual fund manager with a certain amount of capital to invest. You must decide how to allocate that capital among available assets. Your goal is to maximize the fund's expected return. However, the portfolio of investments must be diversified, so as not to be too risky. This example illustrates the main features of an optimization problem.

1. **Decision Variables:** What the decision maker is able to choose or adjust in the decision process. In our mutual fund example, this is the amount of capital to allocate to each available asset.
2. **Objective function:** The performance measure to be optimized (i.e., maximized or minimized) in the decision process. Insofar as the optimization model is concerned, this is the central way of telling whether one set of decisions is “better” than another. In our mutual fund example, the objective function to be maximized is the expected return.
3. **Constraints:** Restrictions imposed on the decision variables. Each constraint represents a *relation* between a left-hand side and a right-hand side. Typically, at least one of these depends on the decision variables, so that the constraint ultimately captures a condition that the decisions must obey. For example, a mutual fund manager cannot invest more than the limited amount of capital at her disposal, so that the capital invested (which depends on the decisions) cannot exceed the available capital. Her decisions may also be subject to requirements related to the level of portfolio diversification and risk of the chosen portfolio.
4. **Parameters:** These constitute inputs in the optimization problem which are typically outside the control of the decision maker. Speaking loosely, the parameters are the “numbers” or “data” of an optimization model. In the investment example, such parameters could be the returns of the various risky assets, the initial capital available for the investment, or the maximum level of risk tolerated in the portfolio.

Optimization is useful in many different settings. This note shows how to formulate and solve a **linear optimization problem** using the Gurobi solver in Python, in the context of a very simple tree planting example.

1.1 Oro Verde Tree Planting

Oro Verde, a non-profit environmental organization, has set up a small tree-planting project on a donated parcel of 150,000 square feet of land, in a remote and deforested area. Oro Verde estimates that the water supply in the region is somewhat limited, with useable reserves not exceeding 50,000 gallons. The project is designed to help large enterprises reduce their carbon footprint, at a minimal cost. In particular, Oro Verde has committed to deliver, precisely 10 years into the future, a total annual sequestration of at least 1800 pounds of carbon. To deliver on the promise, Oro Verde is considering planting three types of trees: maple, elm, and spruce. The trees have very different sequestration rates, require a different amount of water in the early stages of development, and grow to very different widths when they are mature. They also have different 10-year survival rates, and the required seedlings have fairly different prices. Oro Verde estimated that a Maple tree, which would deliver 4.3 pounds of carbon sequestration annually, would require roughly 76 gallons of water and 130 square feet of land. Also, the 10-year survival rate for a maple tree is roughly 70%, and the cost of a seedling is \$5.40. An Elm would deliver an annual sequestration rate of 3.7 pounds, requiring 48 gallons of water and 3,600 square feet; the survival rate would be a bit lower, at 60%, but the price would also be commensurate, at \$3.20 per seedling. Finally, a mature spruce would deliver a sequestration rate of 2.8 pounds, requiring 34 gallons of water and 400 square feet of land; the survival rate would be quite high, 80%, but the cost would also be higher, at \$4.50 per seedling. Oro Verde purchases the seedlings from small communities, which have variable amounts available: currently, only 1,000 seedlings of maple, 500 of elm, and 1,500 of spruce can be purchased. In an effort to stimulate the aging population of elms, Oro Verde has also made a special commitment to purchase and plant at least 15 elm seedlings. Oro Verde is seeking your advice concerning the best course of action...

GENERAL MODELING ADVICE

The first step in building any optimization model is to identify the key ingredients: **the decision variables, the objective, and the constraints**. So before rushing to the keyboard/software, **read the case several times!** You may find it helpful to **highlight certain words** that can help you identify these key ingredients and perhaps also **build an influence diagram**. And it is also very helpful to **write the model on a piece of paper**, before starting to code it! Some of these points will become more clear subsequently...

1.2 The Decision Variables

Oro Verde must choose the number of each type of seedling to plant. We use the notation:

Decision Variables

M = number of **Maple** seedlings planted;
 E = number of **Elm** seedlings planted;
 S = number of **Spruce** seedlings planted.

REMARK. There may be multiple ways to define the decision variables in a given problem. Picking the “right” set is actually a skill that requires significant practice and is often one of the most difficult steps in optimization modeling. A good choice is driven by several considerations: it should capture all the actions under the decision maker’s control, it should allow expressing all

the model ingredients “appropriately,” it should result in a “small-sized” optimization model, etc. Picking the right decision variables may also require some back-and-forth: you might start with a small set of decisions and, when expressing your model, realize that you need to add more.

1.3 The Objective Function

Oro Verde’s goal is to minimize the cost of the project. The **objective function** thus corresponds to the cost of purchasing the seedlings, which is given by the following expression:

$$\text{Cost} = \$5.40 \cdot M + \$3.20 \cdot E + \$4.50 \cdot S. \quad (0)$$

1.4 The Constraints

According to the case, each tree type has a certain width, and thus requires a particular amount of land to mature: 130 square feet for a Maple tree, 3,600 square feet for an Elm tree, and 400 square feet for a Spruce tree. When planting M maples, E elms and S spruce trees, Oro Verde would thus require a total of $130 \cdot M + 3,600 \cdot E + 400 \cdot S$ square feet of land. Since Oro Verde only has 150,000 square feet of land available for the project, since generates a *constraint* on the planting decisions:

$$130 \cdot M + 3,600 \cdot E + 400 \cdot S \leq 150,000 \quad (\text{land}). \quad (1)$$

Constraint

A **constraint** is a relationship between a left-hand-side (LHS) term and a right-hand-side (RHS) term that are compared through a “ $\geq$ ”, “ $\leq$ ” or “ $=$ ” sign.

It is important to note that the same constraint or relationship can be expressed in many ways. For instance, a generic constraint $A \leq B$ could also be expressed as any of the following relations:

$$A - B \leq 0, \quad B - A \geq 0, \quad B \geq A.$$

Although these are all equivalent, we do have a best-practice recommendation.

Best Practice in Writing Constraints

When constructing optimization models, express the constraints so that the **right-hand side corresponds to a problem parameter** (i.e., data). If there is no way to leave a parameter on the RHS, move all the terms to the LHS and leave the RHS as 0.

Note that constraint (1) satisfies this extra requirement. Apart from providing a more principled approach to model-construction, this also gives each constraint a physical interpretation that will make our lives easier later in the course, when we discuss sensitivity analysis.

Continuing with our model, since each tree requires water and only 50,000 gallons of water are available, we have an additional constraint due to water availability:

$$76 \cdot M + 48 \cdot E + 34 \cdot S \leq 50,000 \quad (\text{water}). \quad (2)$$

That the communities can only provide Oro Verde with 1,000 seedlings of maple, 500 of elm, and 1,500 of spruce means that the decision variables must also satisfy the constraints:

$$M \leq 1,000 \quad (3)$$

$$E \leq 500 \quad (4)$$

$$S \leq 1,500. \quad (5)$$

Oro Verde has to plant at least 15 elm trees, so

$$E \geq 15. \quad (6)$$

Oro Verde has promised to deliver 10 years into the future a total annual sequestration of at least 1,800 pounds of carbon. The trees have different 10-year survival rates, of 70%, 60%, and 80% respectively. In addition, a mature tree delivers 4.3, 3.7, and 2.8 pounds of carbon sequestration annually, respectively. Thus, the planting decisions must satisfy the following constraint:

$$0.70 \cdot 4.3 \cdot M + 0.60 \cdot 3.7 \cdot E + 0.80 \cdot 2.80 \cdot S \geq 1,800. \quad (7)$$

Finally, Oro Verde cannot plant a negative number of seedlings, so its decisions must also satisfy:

$$M \geq 0, E \geq 0, S \geq 0. \quad (8)$$

1.5 The Parameters

The parameters in the Oro Verde case are all the “numbers” appearing in the text. For instance, the 150,000 (square feet of available land), the 50,000 (gallons of available water), the \$5.40, \$3.20 and \$4.50 (for the cost of a Maple, Elm and Spruce seedling, respectively), etc.

1.6 The Complete Oro Verde Optimization Problem

In summary, Oro Verde’s decision problem can be expressed compactly as finding M, E, S to:

minimize	$5.40 \cdot M + 3.20 \cdot E + 4.50 \cdot S$	(cost objective)
subject to	$130 \cdot M + 3,600 \cdot E + 400 \cdot S \leq 150,000$	(land constraint) (1)
	$76 \cdot M + 48 \cdot E + 34 \cdot S \leq 50,000$	(water constraint) (2)
	$M \leq 1,000$	(Maple seedling constraint) (3)
	$E \leq 500$	(Elm seedling constraint) (4)
	$S \leq 1,500$	(Spruce seedling constraint) (5)
	$E \geq 15$	(enough Elms constraint) (6)
	$0.70 \cdot 4.3 \cdot M + 0.60 \cdot 3.7 \cdot E + 0.80 \cdot 2.80 \cdot S \geq 1,800$	(sequestration constraint) (7)
	$M \geq 0, E \geq 0, \text{ and } S \geq 0.$	(non-negativity constraints) (8)

2 Linear Optimization and Linear Expressions

Oro Verde's Problem is an example of a **linear optimization problem**.

Linear Optimization Problem

In a **linear optimization problem**, the objective function and the left- and right-hand-sides of all constraints only involve expressions that are **linear in the decision variables**, i.e., that can be written as a sum of terms of the form (coefficient) · (decision variable), where each "coefficient" only depends on problem parameters/data, but not on the decisions.

As the definition and the name suggest, **linearity in decisions** is critical to determining if we are dealing with a linear optimization problem. To understand this, recall from your math courses the concept of a function $f(x)$ that is linear in the variable x . Any such function has the form:

$$f(x) = \text{slope} \cdot x + \text{intercept},$$

where "slope" and "intercept" are two numbers. Expressions in linear optimization are a generalization of this to a case with multiple variables. For instance, in Oro Verde we have three decision variables (M, E, S), and any function f that is **linear in the variables M, E, S** must be of the form

$$f(M, E, S) = \text{coefficient}_M \cdot M + \text{coefficient}_E \cdot E + \text{coefficient}_S \cdot S + \text{coefficient}_0,$$

where "coefficient _{$M, E, S, 0$} " are numbers that do not depend in any way on M, E, S .

A quick inspection of (0)-(8) reveals that **all** the expressions are linear in M, E, S , which exactly makes Oro Verde a **linear** optimization problem.

In contrast, the following expressions would be **non-linear** functions of the decisions M, E, S :

$E \cdot M, E/M$:	the product or ratio of two decision variables
$E^2, 11\sqrt{E} + 23M$:	raising a decision variable to a power different than 1
$\ln(E), \exp(M)$:	logarithm and exponential functions
$\max(E, 5.0), \min(E, M)$:	maximum and minimum
$\text{IF}(M > 5, M, 0.5 \cdot M)$:	IF(.) expressions (written here in "Excel" language).

TO REMEMBER

In a linear model, you should **NEVER** use expressions that involve IF(.), MIN(.), MAX(.), EXP(.), LOG(.), ABS(.) or any other non-linear functions **applied to the decision variables**.

As for the **coefficients** that multiply the decisions in a linear model, these should obviously not depend on the decision variables (because that would make the expressions nonlinear), but they could depend arbitrarily on the **problem parameters**.

TO REMEMBER

In a linear model, the **coefficients** that multiply the decision variables must **NEVER** depend on the decision variables, but **can depend arbitrarily on the problem parameters**.

To better appreciate this point, consider the expression in the left-hand side of constraint (7). The expression is **linear in the decisions M, E, S** . However, the coefficients $0.70 \cdot 4.3, 0.60 \cdot 3.7$

and $0.80 \cdot 2.80$ are actually obtained as **non-linear functions of the parameters**, since they are products of the survival rates and the sequestration rates. More broadly, we could have taken **any function of the case data** and multiply it with a decision and then add any other function of the data multiplied by another decision, and we would still get a linear expression in the decisions.

3 Some More Terminology in Optimization

We conclude our discussion of optimization by introducing some additional useful vocabulary.

3.1 Feasible Solutions, Feasible and Infeasible Problems

Feasible Solutions, Optimal Solutions

A **feasible solution** is a combination of values for the decision variables that satisfies all the constraints. An optimization problem is said to be **feasible** if it admits feasible solutions. In contrast, when no feasible solutions exist the problem is said to be **infeasible**.

An **optimal solution** is a feasible solution that achieves the best possible objective value among all feasible solutions.

As it turns out, the Oro Verde Problem is actually feasible. But if we were to change the requirement of planting at least 15 Elms to at least 1,000 Elms, the problem would no longer be feasible: these 1,000 Elms alone would exceed the 500 Elm seedlings available.

It is generally extremely hard to tell by inspection whether real-world problems are feasible or not; in fact, being able to tell is almost as hard as solving the problem altogether! But extremely powerful algorithms exist that allow solving such decision problems very quickly.

As a side note, an optimization problem may admit multiple optimal solutions; we will learn later how to recognize when that happens, and how to pick among such solutions.

3.2 Binding Constraints and Slack

For a given feasible solution, each constraint can be labeled into one of two categories.

Tight/Binding Constraints and Slack

Given a feasible solution, a constraint is said to be **binding** (or **tight**) if it holds with equality, i.e., if its right-hand-side equals its left-hand-side. If the constraint is not binding, then its **slack** is given by the difference between the right-hand-side and the left-hand-side.

Informally, tight/binding constraints are the ones that matter at that particular solution. In contrast, constraints that are not tight still have some “wiggle room,” and the **slack** exactly measures how much wiggle room exists. Typically, the **slack** is given by the absolute difference between the two sides of the constraint, so that it is always non-negative. In Gurobi, however, the **slack** is calculated as the right-hand-side minus the left-hand-side, so the slack can be both positive (in $\leq$ constraints) or negative (in $\geq$ constraints).

3.3 Unbounded Objectives

An optimization problem is **unbounded** when it is possible to push its objective arbitrarily much in the desired direction. For instance, in Oro Verde this would mean being able to find a feasible planting scheme that generates a cost of $-\infty$. Real-world problems do not admit unbounded objectives, so this is usually a sign that some of the constraints are either missing or incorrect.